Recommender System approach 1:

Collaborative Filtering via GCN over user-item graph and Content-based filtering via DPR embeddings of movie Descriptions

Parameter	Output
Precision@10	0.1560
Recall@10	0.1350
NDCG@10	0.2031
HitRate@10	0.6627

Parameter Name	Value
`SEED`	`42`
`test_size`	`0.2`
`random_state`	`42`
`batch_size`	`16`
`dpr_ctx_model_name`	`"facebook/dpr-ctx_encoder-single-nq-base"`
`max_length`	`128`
`embedding_dim`	`768`
`edge_index`	User–item bidirectional graph edges
`data`	`torch_geometric.data.Data` graph object
`dropout`	`0.3`
`freeze_items`	`False`
`dim`	`768`
`item_adapter`	`nn.Linear(768, 768, bias=False)`
`gcn1`	`GCNConv(768, 768)`
`gcn2`	`GCNConv(768, 768)`
`activation`	`ReLU`
`optimizer`	`Adam`
`learning_rate`	`1e-3`
`EPOCHS`	`100`
`L2_W`	`1e-6`
`loss_function`	Bayesian Personalized Ranking (BPR) loss
`regularization`	L2 norm regularization
`negative_samples_per_user`	`1`
evaluation_k	10

Recommender System approach 2:

Hybrid recommender that learns user preferences from graph structure (via GCN) while enriching movies with semantic text embeddings (via a pretrained SimCSE model)

Parameter	Output
Precision@10	0.1266
Recall@10	0.1396
NDCG@10	0.1724
HitRate@10	0.6442

Parameter Name	Value
`SEED`	`42`
`Positive rating threshold`	`>= 4.0`
`SimCSE model`	`"princeton-nlp/sup-simcse-bert-base-uncased"`
`SimCSE batch size`	`32`
`Embedding dimension`	`768` (from SimCSE output)
`Train/Test split ratio`	`80% / 20%`
`Graph type`	`User-Item Bipartite Graph`
`GCN layers`	`2`
`GCN hidden dimension`	`Same as embedding_dim (768)`
`Dropout`	`0.3`
`Freeze item embeddings`	`False`
`Optimizer`	`Adam`
`Learning rate (lr)`	`1e-3`
`Regularization weight (L2_W)`	`1e-6`
`Epochs`	`10000`
`Loss function`	`Bayesian Personalized Ranking (BPR)`
`Device`	`"cuda:6"` if available, else `"cpu"`
`Evaluation metric K`	`10`
`Evaluation metrics`	`Precision@10`, `Recall@10`, `NDCG@10`, `HitRate@10`
`Cold-start encoding model`	`SimCSE SentenceTransformer`
`Negative sampling`	`1 random negative item per user per epoch`

When Epoch is 100

Parameter	Output
Precision@10:	0.1595
Recall@10:	0.1458
NDCG@10:	0.2088

Parameter	Output
HitRate@10:	0.6813

Recommender System Approach 3

content-aware GCN recommender that learns from user–item interactions and enriches movie nodes with INSTRUCTOR embeddings

Parameter Name	Value
Random Seed	`42`
Positive Rating Threshold	`>= 4.0`
Feedback Type	Implicit (binary: liked = 1)
Text Embedding Model	`hkunlp/instructor-base`
Instruction for Embedding	`"Represent the movie for recommendation:"`
Embedding Batch Size	`32`
Embedding Dimension	`768`
Graph Type	User–Item Bipartite Graph
GCN Layers	`2` (`GCNConv`)
Hidden Dimension	Same as embedding_dim (`768`)
Dropout Rate	`0.3`
Freeze Item Embeddings	`False`
Optimizer	`Adam`
Learning Rate	`1e-3`
L2 Regularization Weight (`L2_W`)	`1e-6`
Loss Function	Bayesian Personalized Ranking (BPR)
Negative Sampling Strategy	1 random negative item per user per epoch
Train/Test Split Ratio	`80% / 20%`
Number of Training Epochs	`10000`
Evaluation Metric K	`10`

Parameter	Output
Precision@10:	0.1275
Recall@10:	0.1367
NDCG@10:	0.1714
HitRate@10:	0.6341

When Epoch is 100

Parameter	Output
Precision@10:	0.1278
Recall@10:	0.1066
NDCG@10:	0.1588
HitRate@10:	0.6155

Recommender System Approach 4

Content-aware graph recommender where BGE embeddings give semantic meaning to movies, GCN layers propagate collaborative signals and BPR loss optimizes personalized rankings

Parameter Name	Value
Random Seed	`42`
Positive Rating Threshold	`>= 4.0`
Feedback Type	Implicit (binary)
Text Embedding Model	`"BAAI/bge-base-en-v1.5"`
Embedding Batch Size	`32`
Embedding Dimension	`768`
Graph Type	User–Item Bipartite
GCN Layers	`2` (`GCNConv`)
Dropout Rate	`0.3`
Hidden Dimension	`768`
Freeze Item Embeddings	`False`
Optimizer	`Adam`
Learning Rate	`1e-3`
Weight Decay	`0.0`
L2 Regularization Weight (`L2_W`)	`1e-6`
Loss Function	Bayesian Personalized Ranking (BPR)
Negative Sampling Strategy	1 random item per user per epoch
Train/Test Split Ratio	`80% / 20%`
Training Epochs	`10000`

Parameter	Output
Precision@10:	0.1250
Recall@10:	0.1317
NDCG@10:	0.1662
HitRate@10:	0.6256

Parameter	Output
Precision@10:	0.1384
Recall@10:	0.1192
NDCG@10:	0.1727
HitRate@10:	0.6408

Recommender System Approach 5

Graph-based collaborative filtering (GCN) with textual embeddings from MiniLM

Parameter Name	Value
`SEED`	`42`
`model_name`	`"all-MiniLM-L6-v2"`
`batch_size`	`32`
`embedding_dim`	`384`
`test_size`	`0.2`
`dropout`	`0.3`
`gcn_hidden`	`None`
`freeze_items`	`False`
`lr`	`1e-3`
`EPOCHS`	`10000`
`L2_W`	`1e-6`
`k`	`10`
`rating_threshold`	`4.0`
`neg_sampling_method`	`Random uniform`
`bpr_loss`	`BPR`
`edge_index`	`User–Item bidirectional`
`train_df`	`80% of data`
`test_df`	`20% of data`
`optimizer`	`Adam`
`precision_recall_ndcg_hit.k`	`10`
`minilm_vec`	`MiniLM encoded text`
`item_adapter`	`Linear(dim, dim)`
`user_embedding`	`num_users × dim`
`item_embedding`	`num_items × dim`
`train_df_tensor`	`torch.tensor(train_df[['userId','movieId']])`
`num_edges`	`2 × len(train_df)`

Parameter	Output
Precision@10:	0.1366
Recall@10:	0.1393

Parameter	Output
NDCG@10:	0.1828
HitRate@10:	0.6796

100 Epochs

Parameter	Output
Precision@10:	0.1378
Recall@10:	0.1204
NDCG@10:	0.1736
HitRate@10:	0.6442